

B9 Transformations

Name: _____

Class: _____

Date: _____

Time: **401 minutes**

Marks: **336 marks**

Comments:

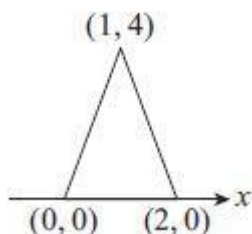
EXAM PAPERS PRACTICE

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Q1.

Figure 1 shows $y = f(x)$.

Figure 1



Which figure below shows $y = f(2x)$?

Tick **one** box.

Figure 2

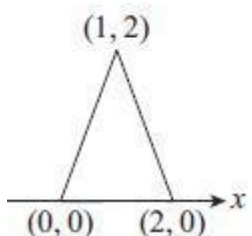


Figure 3

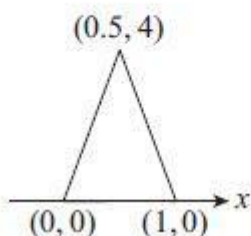


Figure 4

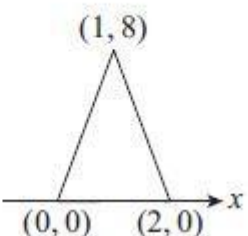


Figure 5

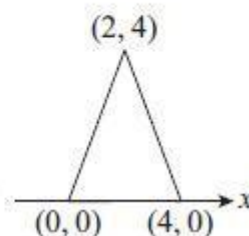


Figure 2

☐

Figure 3

☐

Figure 4

☐

Figure 5

☐

(Total 1 mark)

Q2.

The graph of $y = 5^x$ is transformed by a stretch in the y -direction, scale factor 5

State the equation of the transformed graph.

Circle your answer.

$$y = 5 \times 5^x$$

$$y = 5^{\frac{x}{5}}$$

$$y = \frac{1}{5} \times 5^x$$

$$y = 5^{5x}$$

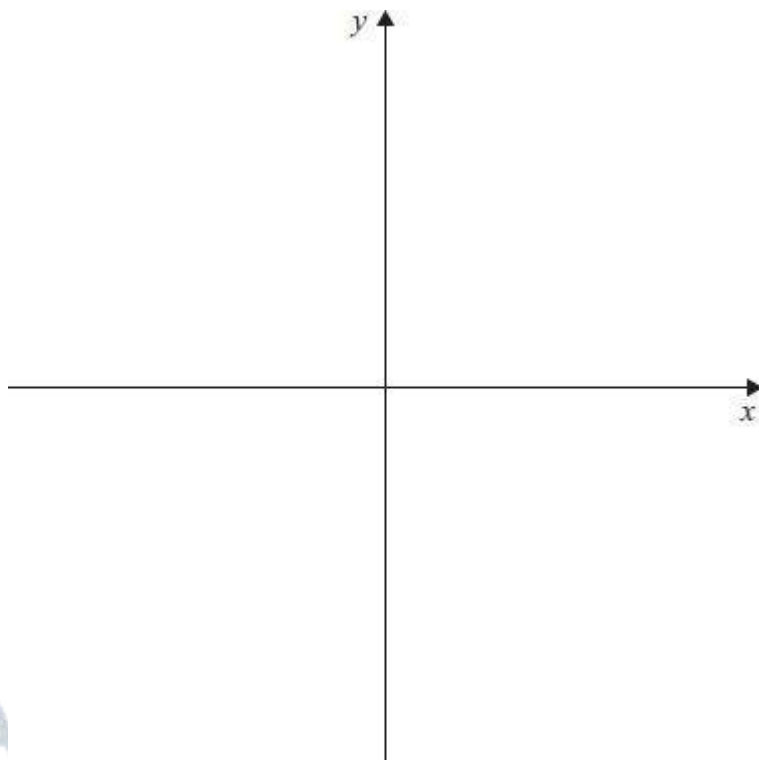
(Total 1 mark)

Q3.

A curve, C , has equation $y = x^2 - 6x + k$, where k is a constant.

The equation $x^2 - 6x + k = 0$ has two distinct positive roots.

(a) Sketch C on the axes below.



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(2)

(b) Find the range of possible values for k .

Fully justify your answer.

(4)

(Total 6 marks)

Q4.

(a) Determine a sequence of transformations which maps the graph of $y = \sin x$ onto the graph of $y = \sqrt{3} \sin x - 3 \cos x + 4$

Fully justify your answer.

(7)

- (b) (i) Show that the least value of $\frac{1}{\sqrt{3} \sin x - 3 \cos x + 4}$ is $\frac{2 - \sqrt{3}}{2}$ (2)

- (ii) Find the greatest value of $\frac{1}{\sqrt{3} \sin x - 3 \cos x + 4}$ (1)
(Total 10 marks)

Q5.

The curve $y = \sqrt{x}$ is translated onto the curve $y = \sqrt{x+4}$

The translation is described by a vector.
Find this vector.
Circle your answer.

$$\begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 4 \end{bmatrix}$$

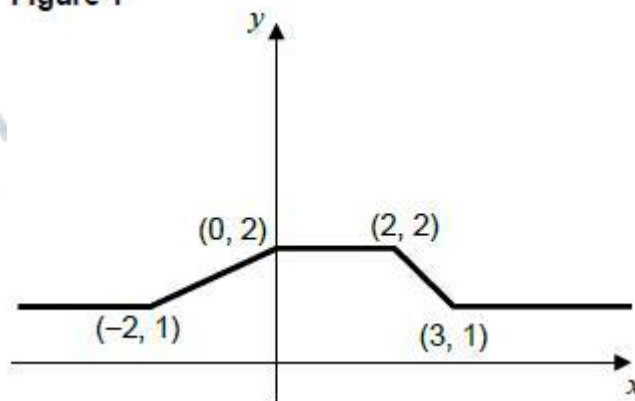
$$\begin{bmatrix} 0 \\ -4 \end{bmatrix}$$

(Total 1 mark)

Q6.

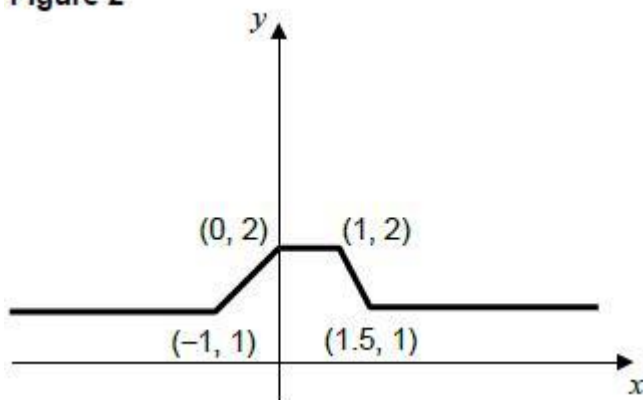
The graph of $y = f(x)$ is shown in **Figure 1**.

Figure 1



State the equation of the graph shown in **Figure 2**.

Figure 2



Circle your answer.

$y = f(2x)$
 $y = f\left(\frac{x}{2}\right)$
 $y = 2f(x)$
 $y = \frac{1}{2}f(x)$

(Total 1 mark)

Q7.

- (a) Determine a sequence of transformations which maps the graph of $y = \cos\theta$ onto the graph of $y = 3\cos\theta + 3\sin\theta$

Fully justify your answer.

(6)

- (b) Hence or otherwise find the least value and greatest value of

$$4 + (3\cos\theta + 3\sin\theta)^2$$

Fully justify your answer.

(3)

(Total 9 marks)

Q8.

- (a) (i) Express $x^2 - 6x + 11$ in the form $(x - p)^2 + q$.

(2)

- (ii) **Use the result from part (a)(i)** to show that the equation $x^2 - 6x + 11 = 0$ has no real solutions.

(2)

- (b) A curve has equation $y = x^2 - 6x + 11$.

- (i) Find the coordinates of the vertex of the curve.

(2)

- (ii) Sketch the curve, indicating the value of y where the curve crosses the y -axis. (3)
- (iii) Describe the geometrical transformation that maps the curve with equation $y = x^2 - 6x + 11$ onto the curve with equation $y = x^2$. (3)
- (Total 12 marks)

Q9.

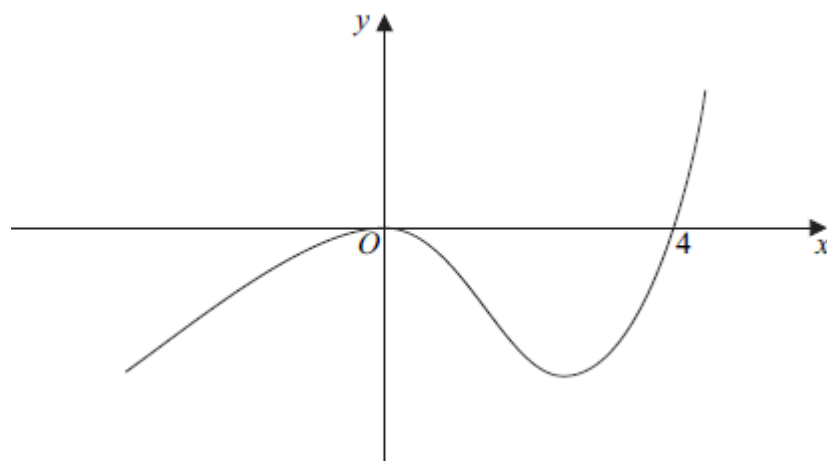
- (a) Describe a geometrical transformation that maps the graph of $y = 4^x$ onto the graph of $y = 3 \times 4^x$. (2)
- (b) Sketch the curve with equation $y = 3 \times 4^x$, indicating the value of the intercept on the y -axis. (2)
- (a) The curve with equation $y = 4^{-x}$ intersects the curve $y = 3 \times 4^x$ at the point P . Use logarithms to find the x -coordinate of P , giving your answer to three significant figures. (5)
- (Total 9 marks)

Q10.

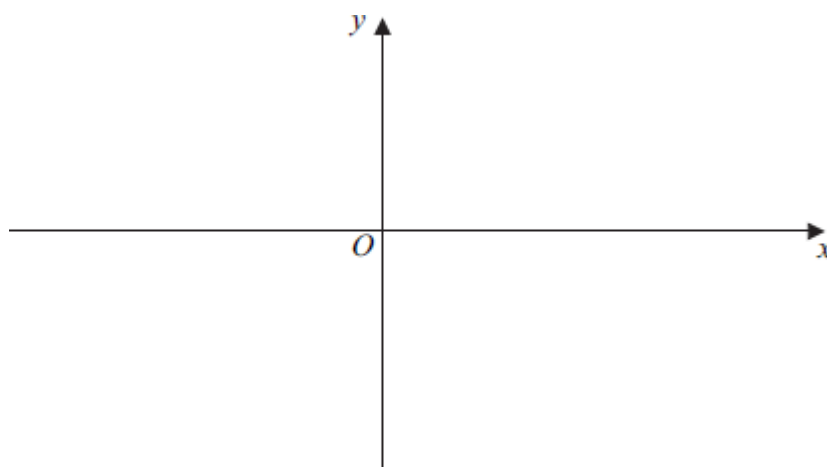
- (a) Write down the two solutions of the equation $\tan(x + 30^\circ) = \tan 79^\circ$ in the interval $0^\circ \leq x \leq 360^\circ$. (2)
- (b) Describe a single geometrical transformation that maps the graph of $y = \tan x$ onto the graph of $y = \tan(x + 30^\circ)$. (2)
- (c) (i) Given that $5 + \sin^2 \theta = (5 + 3 \cos \theta) \cos \theta$, show that $\cos \theta = \frac{3}{4}$. (5)
- (ii) **Hence** solve the equation $5 + \sin^2 2x = (5 + 3 \cos 2x) \cos 2x$ in the interval $0 < x < 2\pi$, giving your values of x in radians to three significant figures. (3)
- (Total 12 marks)

Q11.

The diagram shows a sketch of the curve with equation $y = f(x)$.



- (a) On the axes below, sketch the curve with equation $y = |f(x)|$.



(2)

- (b) Describe a sequence of two geometrical transformations that maps the graph of $y = f(x)$ onto the graph of $y = f(2x - 1)$.

(4)

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(Total 6 marks)

Q12.

A circle C has equation

$$x^2 + y^2 - 10x + 14y + 25 = 0$$

- (a) Write the equation of C in the form

$$(x - a)^2 + (y - b)^2 = k$$

where a , b and k are integers.

(3)

- (b) Hence, for the circle C , write down:

- (i) the coordinates of its centre; (1)
- (ii) its radius. (1)
- (c) (i) Sketch the circle C . (2)
- (ii) Write down the coordinates of the point on C that is furthest away from the x -axis. (2)
- (d) Given that k has the same value as in part (a), describe geometrically the transformation which maps the circle with equation $(x + 1)^2 + y^2 = k$ onto the circle C . (3)
- (Total 12 marks)**

Q13.

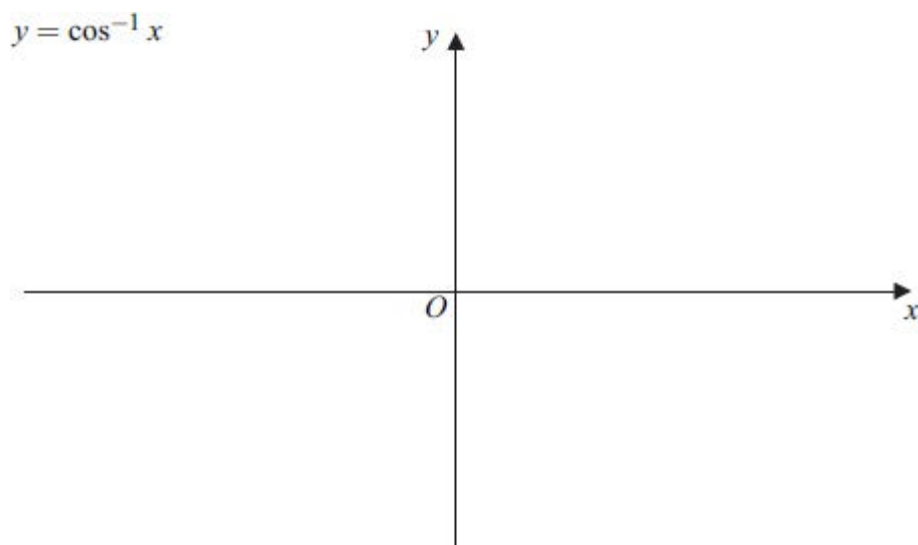
- (a) Sketch the graph of $y = 9^x$, indicating the value of the intercept on the y -axis. (2)
- (b) Use logarithms to solve the equation $9^x = 15$, giving your value of x to three significant figures. (2)
- (c) The curve $y = 9^x$ is reflected in the y -axis to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (1)
- (Total 5 marks)**

Q14.

- (a) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 \sqrt{8x^3 + 1} \, dx$, giving your answer to three significant figures. (4)
- (b) Describe the single transformation that maps the graph of $y = \sqrt{8x^3 + 1}$ onto the graph of $y = \sqrt{x^3 + 1}$. (2)
- (c) The curve with equation $y = \sqrt{x^3 + 1}$ is translated by $\begin{bmatrix} 2 \\ -0.7 \end{bmatrix}$ to give the curve with equation $y = g(x)$. Find the value of $g(4)$. (3)
- (Total 9 marks)**

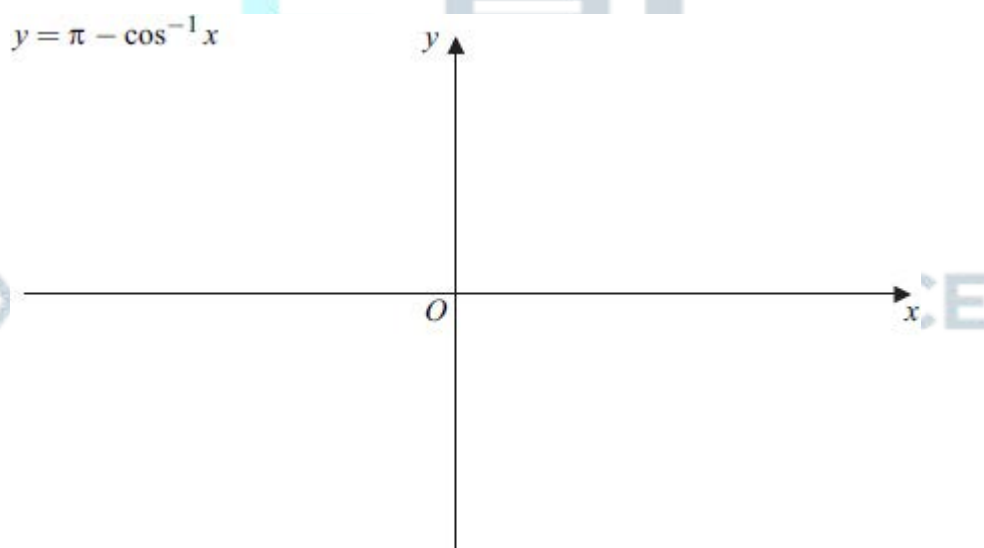
Q15.

- (a) Sketch the graph of $y = \cos^{-1} x$, where y is in radians. State the coordinates of the end points of the graph.



(2)

- (b) Sketch the graph of $y = \pi - \cos^{-1} x$, where y is in radians. State the coordinates of the end points of the graph.

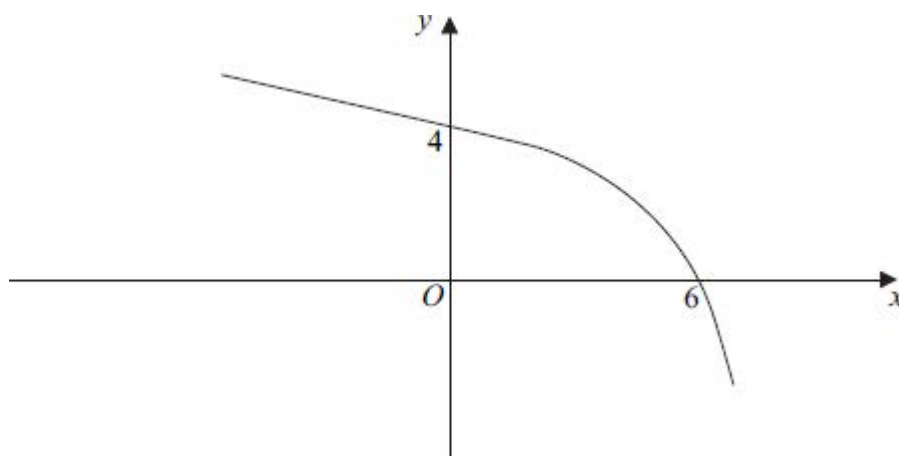


(2)

(Total 4 marks)

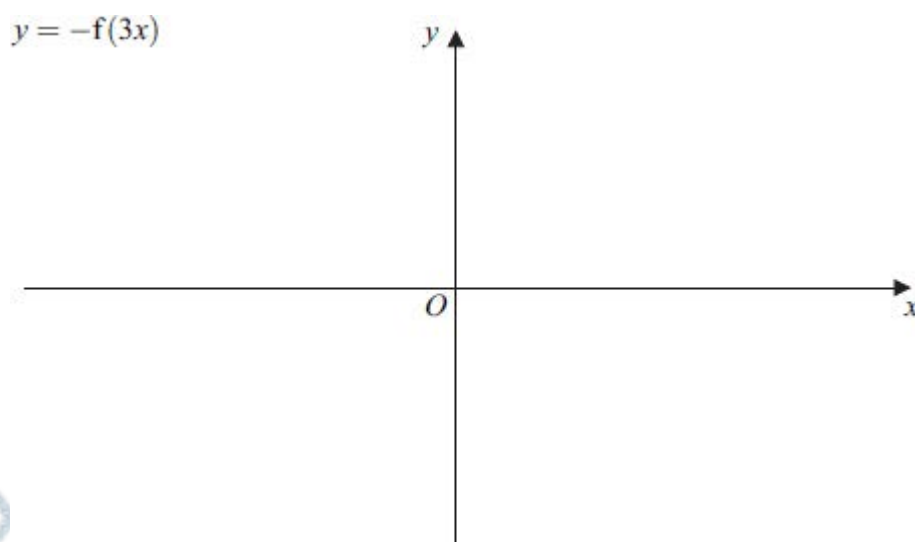
Q16.

The diagram shows a sketch of the curve with equation $y = f(x)$.



- (a) On **Figure 1**, below, sketch the curve with equation $y = -f(3x)$, indicating the values where the curve cuts the coordinate axes.

Figure 1

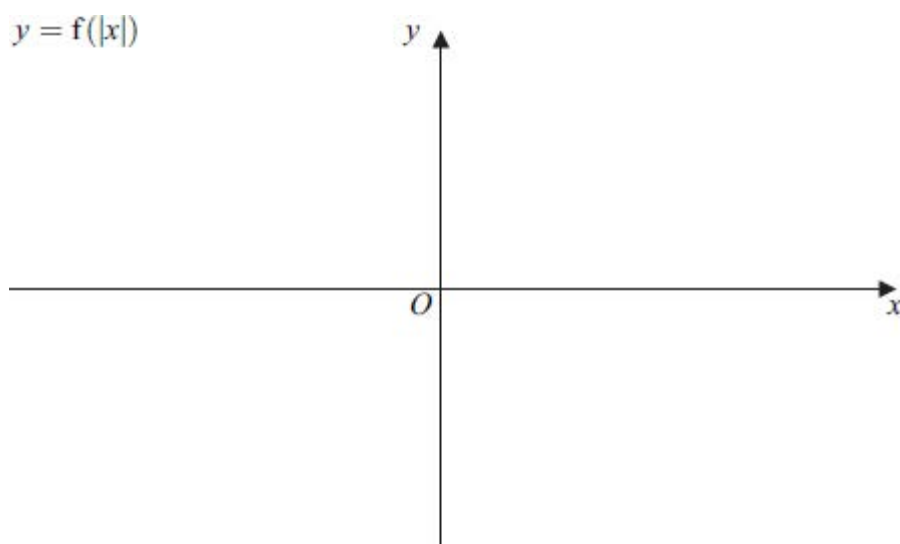


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(2)

- (b) On **Figure 2**, on the opposite page, sketch the curve with equation $y = f(|x|)$, indicating the values where the curve cuts the coordinate axes.

Figure 2



(3)

- (c) Describe a sequence of two geometrical transformations that maps the graph of $y = f(x)$ onto the graph of $y = f\left(-\frac{1}{2}x\right)$.

(4)

(Total 9 marks)

Q17.

- (a) Factorise $x^2 - 4x - 12$.
- (b) Sketch the graph with equation $y = x^2 - 4x - 12$, stating the values where the curve crosses the coordinate axes.

(1)

(4)

- (c) (i) Express $x^2 - 4x - 12$ in the form $(x - p)^2 - q$, where p and q are positive integers.

(2)

- (ii) Hence find the minimum value of $x^2 - 4x - 12$.

(1)

- (d) The curve with equation $y = x^2 - 4x - 12$ is translated by the vector $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$.

Find an equation of the new curve. You need not simplify your answer.

(2)

(Total 10 marks)

Q18.

- (a) (i) Describe the geometrical transformation that maps the graph of

$y = \left(1 + \frac{x}{3}\right)^6$ onto the graph of $y = (1 + 2x)^6$.

(2)

- (ii) The curve $y = \left(1 + \frac{x}{3}\right)^6$ is translated by the vector $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$ to give the curve $y = g(x)$.

Find an expression for $g(x)$, simplifying your answer.

(2)

- (b) The first four terms in the binomial expansion of $\left(1 + \frac{x}{3}\right)^6$ are $1 + ax + bx^2 + cx^3$.

Find the values of the constants a , b and c , giving your answers in their simplest form.

(4)

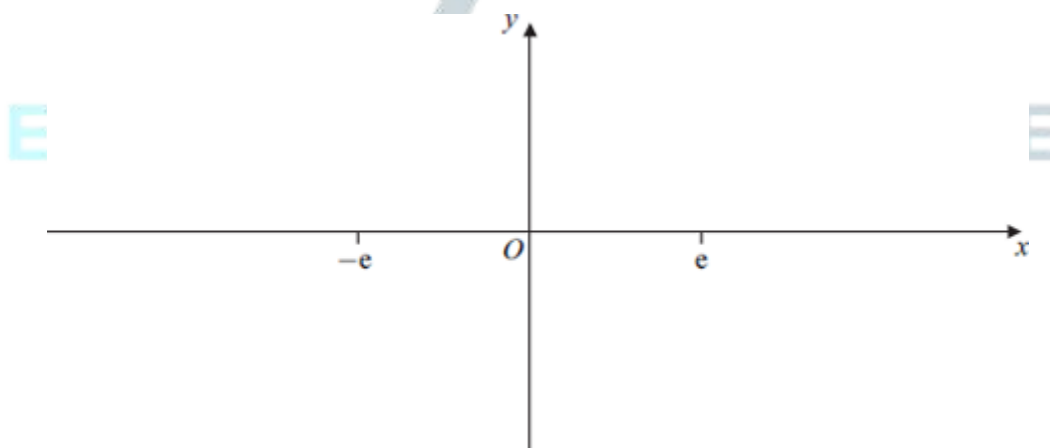
(Total 8 marks)

Q19.

- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = \ln x$ onto the graph of $y = 4 \ln(x - e)$.

(4)

- (b) Sketch, on the axes given below, the graph of $y = |4 \ln(x - e)|$, indicating the exact value of the x -coordinate where the curve meets the x -axis.



(3)

- (c) (i) Solve the equation $|4 \ln(x - e)| = 4$.

(3)

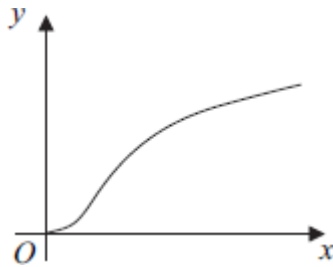
- (ii) Hence, or otherwise, solve the inequality $|4 \ln(x - e)| \geq 4$.

(3)

(Total 13 marks)

Q20.

The diagram shows part of a curve whose equation is $y = \log_{10}(x^2 + 1)$.



- (a) Use the trapezium rule with five ordinates (four strips) to find an approximate value for

$$\int_0^1 (\log_{10}(x^2 + 1)) \, dx$$

giving your answer to three significant figures.

(4)

- (b) The graph of $y = 2 \log_{10} x$ can be transformed into the graph of $y = 1 + 2 \log_{10} x$ by means of a translation. Write down the vector of the translation.

(1)

- (c) (i) Show that $\log_{10}(10x^2) = 1 + 2 \log_{10} x$.

(2)

- (ii) Show that the graph of $y = 2 \log_{10} x$ can also be transformed into the graph of $y = 1 + 2 \log_{10} x$ by means of a **stretch**, and describe the stretch.

(4)

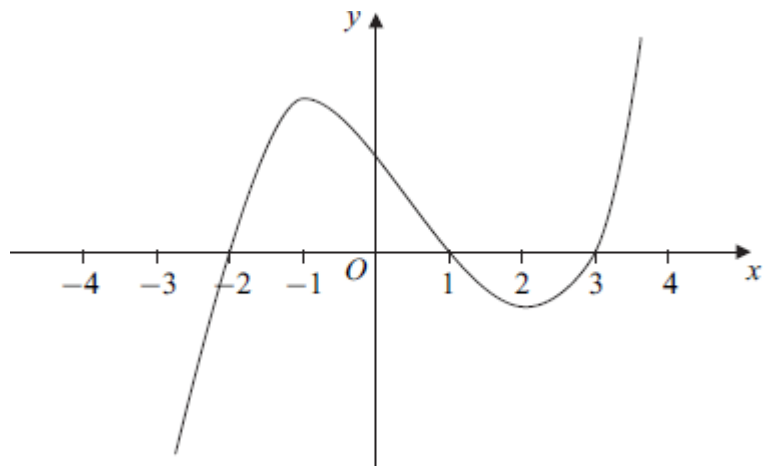
- (iii) The curve with equation $y = 1 + 2 \log_{10} x$ intersects the curve $y = \log_{10}(x^2 + 1)$ at the point P . Given that the x -coordinate of P is positive, find the gradient of the line OP , where O is the origin. Give your answer in the form $\log_{10} \left(\frac{a}{b} \right)$, where a and b are integers.

(4)

(Total 15 marks)

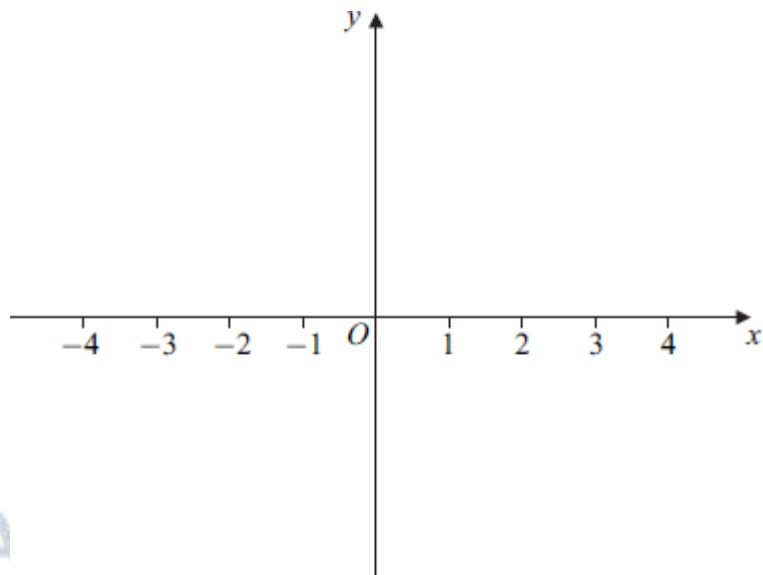
Q21.

The sketch shows part of the curve with equation $y = f(x)$.



- (a) On **Figure 1** below, sketch the curve with equation $y = |f(x)|$.

Figure 1

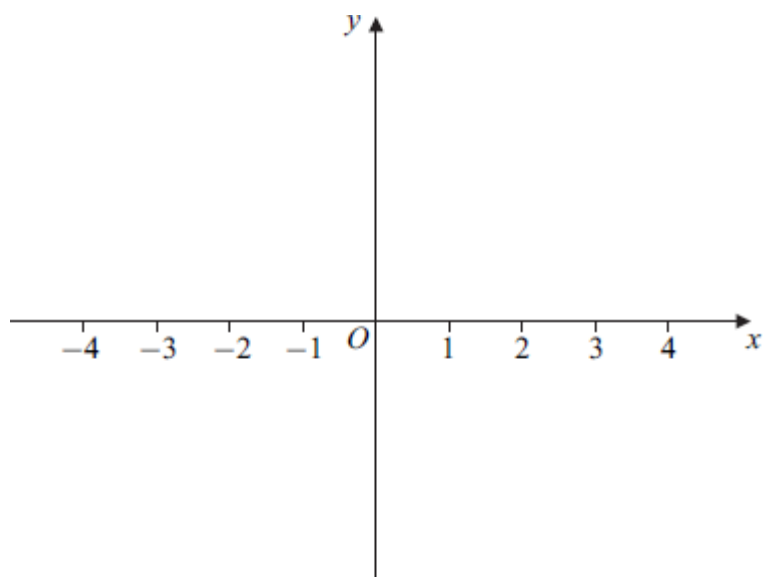


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(3)

- (b) On **Figure 2**, sketch the curve with equation $y = f(|x|)$.

Figure 2



(2)

- (c) Describe a sequence of two geometrical transformations that maps the graph of $y = f(x)$ onto the graph of $y = \frac{1}{2} f(x + 1)$.

(4)

- (d) The maximum point of the curve with equation $y = f(x)$ has coordinates $(-1, 10)$. Find the coordinates of the maximum point of the curve with equation $y = \frac{1}{2} f(x + 1)$.

(2)

(Total 11 marks)

Q22.

- (a) Use the trapezium rule with four ordinates (three strips) to find an approximate value for $\int_0^{1.5} \sqrt{27x^3 + 4} dx$, giving your answer to three significant figures.

(4)

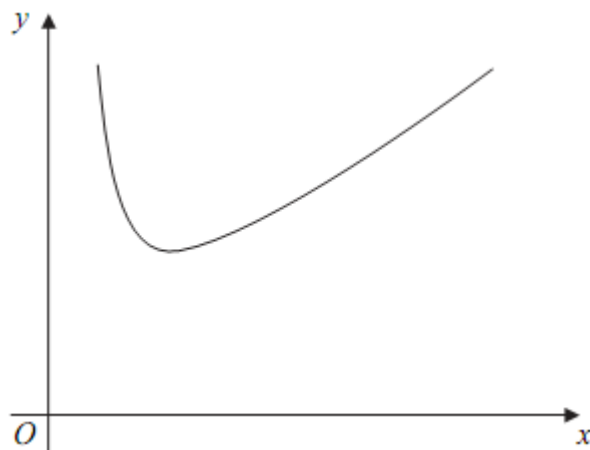
- (b) The curve with equation $y = \sqrt{27x^3 + 4}$ is stretched parallel to the x -axis with scale factor 3 to give the curve with equation $y = g(x)$. Write down an expression for $g(x)$.

(2)

(Total 6 marks)

Q23.

A curve C is defined for $x > 0$ by the equation $y = x + 3 + \frac{8}{x^4}$ and is sketched below.



- (a) Given that $y = x + 3 + \frac{8}{x^4}$, find $\frac{dy}{dx}$. (3)
- (b) Find an equation of the tangent at the point on the curve C where $x = 1$. (3)
- (c) The curve C has a minimum point M . Find the coordinates of M . (4)
- (d) (i) Find $\int \left(x + 3 + \frac{8}{x^4} \right) dx$. (3)
- (ii) Hence find the area of the region bounded by the curve C , the x -axis and the lines $x = 1$ and $x = 2$. (2)
- (e) The curve C is translated by $\begin{bmatrix} 0 \\ k \end{bmatrix}$ to give the curve $y = f(x)$. Given that the x -axis is a tangent to the curve $y = f(x)$, state the value of the constant k . (1)
- (Total 16 marks)**

Q24.

The functions f and g are defined with their respective domains by

$$f(x) = 3 \cos \frac{1}{2}x, \quad \text{for } 0 \leq x \leq 2\pi$$

$$g(x) = |x|, \quad \text{for all real values of } x$$

- (a) Find the range of f . (2)
- (b) The inverse of f is f^{-1} .
- (i) Find $f^{-1}(x)$. (3)
- (ii) Solve the equation $f^{-1}(x) = 1$, giving your answer in an exact form. (2)
- (c) (i) Write down an expression for $gf(x)$. (1)
- (ii) Sketch the graph of $y = gf(x)$ for $0 \leq x \leq 2\pi$. (3)
- (d) Describe a sequence of two geometrical transformations that maps the graph of $y = \cos x$ onto the graph of $y = 3 \cos \frac{1}{2}x$. (3)
- (Total 14 marks)

Q25.

- (a) Express $x^2 + 5x + 7$ in the form $(x + p)^2 + q$, where p and q are rational numbers. (3)
- (b) A curve has equation $y = x^2 + 5x + 7$.
- (i) Find the coordinates of the vertex of the curve. (2)
- (ii) State the equation of the line of symmetry of the curve. (1)
- (iii) Sketch the curve, stating the value of the intercept on the y -axis. (3)
- (c) Describe the geometrical transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 + 5x + 7$. (3)
- (Total 12 marks)

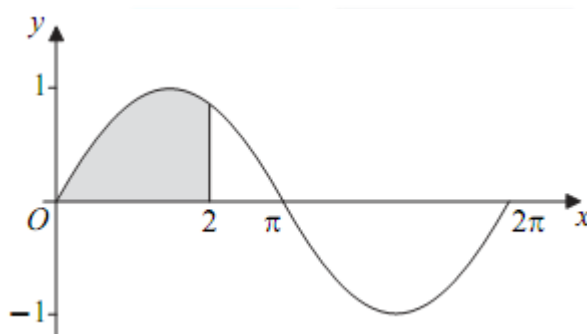
Q26.

- (a) Sketch the curve with equation $y = 4^x$, indicating the coordinates of any point where the curve intersects the coordinate axes. (2)

- (b) Describe the geometrical transformation that maps the graph of $y = 4^x$ onto the graph of $y = 4^x - 5$. (2)
- (c) (i) Use the substitution $Y = 2^x$ to show that the equation $4^x - 2^{x+2} - 5 = 0$ can be written as $Y^2 - 4Y - 5 = 0$. (2)
- (ii) Hence show that the equation $4^x - 2^{x+2} - 5 = 0$ has only one real solution.
- Use logarithms to find this solution, giving your answer to three decimal places. (4)
- (Total 10 marks)

Q27.

A curve C , defined for $0 \leq x \leq 2\pi$ by the equation $y = \sin x$, where x is in radians, is sketched below. The region bounded by the curve C , the x -axis from 0 to 2 and the line $x = 2$ is shaded.



- (a) The area of the shaded region is given by $\int_0^2 \sin x \, dx$, where x is in radians. (4)
- Use the trapezium rule with five ordinates (four strips) to find an approximate value for the area of the shaded region, giving your answer to three significant figures. (4)
- (b) Describe the geometrical transformation that maps the graph of $y = \sin x$ onto the graph of $y = 2 \sin x$. (2)
- (c) Use a trigonometrical identity to solve the equation
- $$2 \sin x = \cos x$$
- in the interval $0 \leq x \leq 2\pi$, giving your solutions in radians to three significant figures. (4)
- (Total 10 marks)

Q28.

- (a) (i) Solve the equation $\operatorname{cosec} \theta = -4$ for $0^\circ < \theta < 360^\circ$, giving your answers to the nearest 0.1° .

(2)

- (ii) Solve the equation

$$2 \cot^2(2x + 30^\circ) = 2 - 7 \operatorname{cosec}(2x + 30^\circ)$$

for $0^\circ < x < 180^\circ$, giving your answers to the nearest 0.1° .

(6)

- (b) Describe a sequence of two geometrical transformations that maps the graph of $y = \operatorname{cosec} x$ onto the graph of $y = \operatorname{cosec}(2x + 30^\circ)$.

(4)

(Total 12 marks)

Q29.

- (a) Express $(x - 5)(x - 3) + 2$ in the form $(x - p)^2 + q$, where p and q are integers.

(3)

- (b) (i) Sketch the graph of $y = (x - 5)(x - 3) + 2$, stating the coordinates of the minimum point and the point where the graph crosses the y -axis.

(3)

- (ii) Write down an equation of the tangent to the graph of $y = (x - 5)(x - 3) + 2$ at its vertex.

(2)

- (c) Describe the geometrical transformation that maps the graph of $y = x^2$ onto the graph of $y = (x - 5)(x - 3) + 2$.

(3)

(Total 11 marks)

Q30.

- (a) Sketch the curve with equation $y = 2^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 2^x \, dx$, giving your answer to three significant figures.

(4)

- (ii) State how you could obtain a better approximation to the value of the integral using the trapezium rule.

(1)

- (c) Describe a geometrical transformation that maps the graph of $y = 2^x$ onto the graph of $y = 2^{x+7} + 3$.

(3)

- (d) The curve $y = 2^{x+k} + 3$ intersects the y -axis at the point $A(0, 8)$.

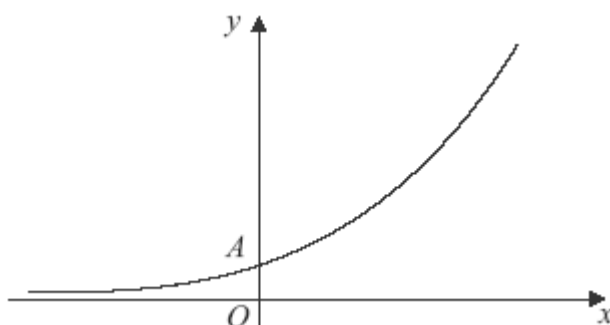
Show that $k = \log_m n$, where m and n are integers.

(2)

(Total 12 marks)

Q31.

The diagram shows a sketch of the curve $y = 2^{4x}$.



The curve intersects the y -axis at the point A .

- (a) Find the value of the y -coordinate of A .

(1)

- (b) Use the trapezium rule with six ordinates (five strips) to find an approximate value for $\int_0^1 2^{4x} dx$, giving your answer to two decimal places.

(4)

- (c) Describe the geometrical transformation that maps the graph of $y = 2^{4x}$ onto the graph of $y = 2^{4x-3}$.

(2)

- (d) The curve $y = 2^{4x}$ is translated by the vector $\begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix}$ to give the curve $y = g(x)$.

The curve $y = g(x)$ crosses the x -axis at the point Q . Find the x -coordinate of Q .

(4)

- (e) (i) Given that

$$\log_a k = 3 \log_a 2 + \log_a 5 - \log_a 4$$

show that $k = 10$.

(3)

- (ii) The line $y = \frac{5}{4}$ crosses the curve $y = 2^{4x-3}$ at the point P . Show that the

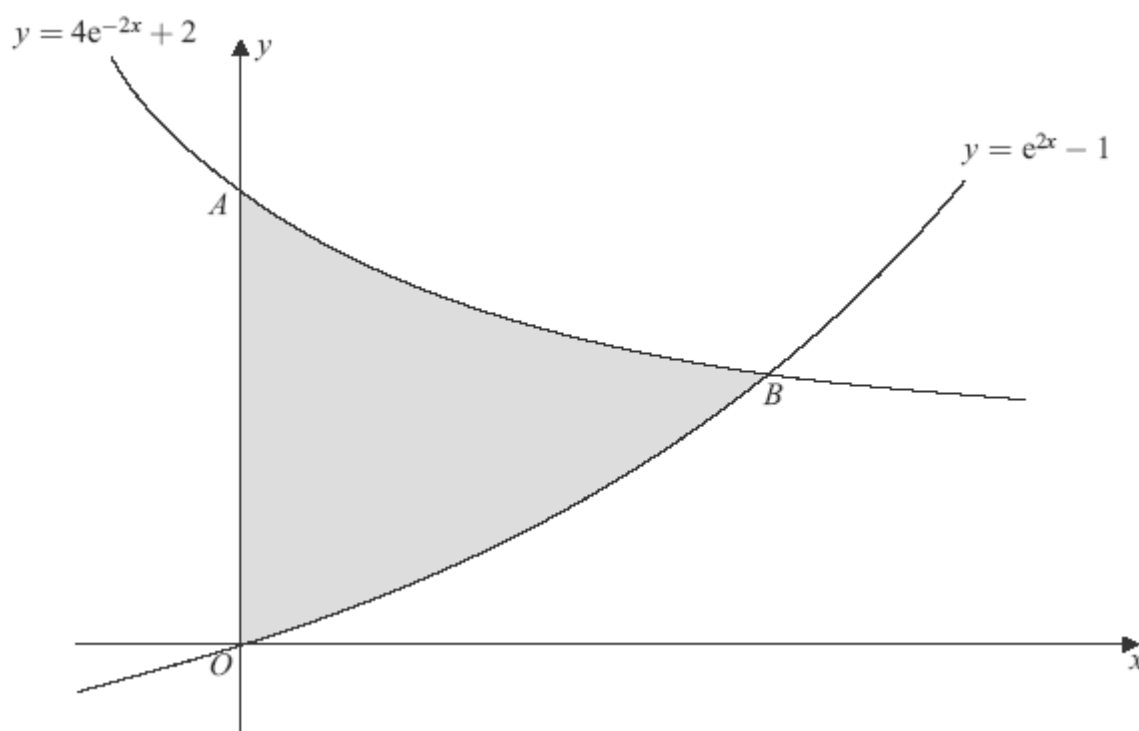
x -coordinate of P is $\frac{1}{4 \log_{10} 2}$.

(3)

(Total 17 marks)

Q32.

The diagram shows the curves $y = e^{2x} - 1$ and $y = 4e^{-2x} + 2$.



The curve $y = 4e^{-2x} + 2$ crosses the y -axis at the point A and the curves intersect at the point B .

- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = e^x$ onto the graph of $y = e^{2x} - 1$.

(4)

- (b) Write down the coordinates of the point A .

(1)

- (c) (i) Show that the x -coordinate of the point B satisfies the equation

$$(e^{2x})^2 - 3e^{2x} - 4 = 0$$

(2)

(ii) Hence find the exact value of the x -coordinate of the point B .

(3)

- (d) Find the exact value of the area of the shaded region bounded by the curves $y = e^{2x} - 1$ and $y = 4e^{-2x} + 2$ and the y -axis.

(5)

(Total 15 marks)

Q33.

- (a) (i) Express $x^2 + 2x + 5$ in the form $(x + p)^2 + q$, where p and q are integers.

(2)

(ii) Hence show that $x^2 + 2x + 5$ is always positive.

(1)

- (b) A curve has equation $y = x^2 + 2x + 5$.

(i) Write down the coordinates of the minimum point of the curve.

(2)

(ii) Sketch the curve, showing the value of the intercept on the y -axis.

(2)

- (c) Describe the geometrical transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 + 2x + 5$.

(3)

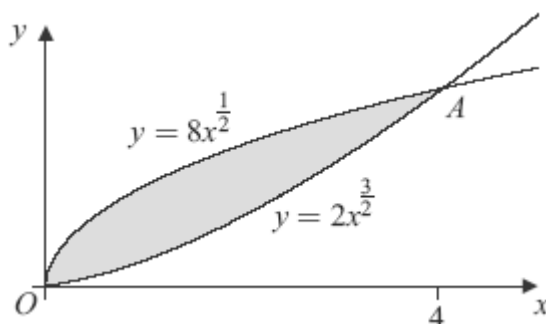
(Total 10 marks)

Q34.

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The diagram shows a sketch of the curves with equations $y = 2x^{\frac{3}{2}}$ and $y = 8x^{\frac{1}{2}}$.



The curves intersect at the origin and at the point A , where $x = 4$.

- (a) (i) For the curve $y = 2x^{\frac{3}{2}}$, find the value of $\frac{dy}{dx}$ when $x = 4$.

(2)

(ii) Find an equation of the normal to the curve $y = 2x^{\frac{3}{2}}$ at the point A .

(4)

(b) (i) Find $\int 8x^{\frac{1}{2}} dx$.

(2)

(ii) Find the area of the shaded region bounded by the two curves.

(4)

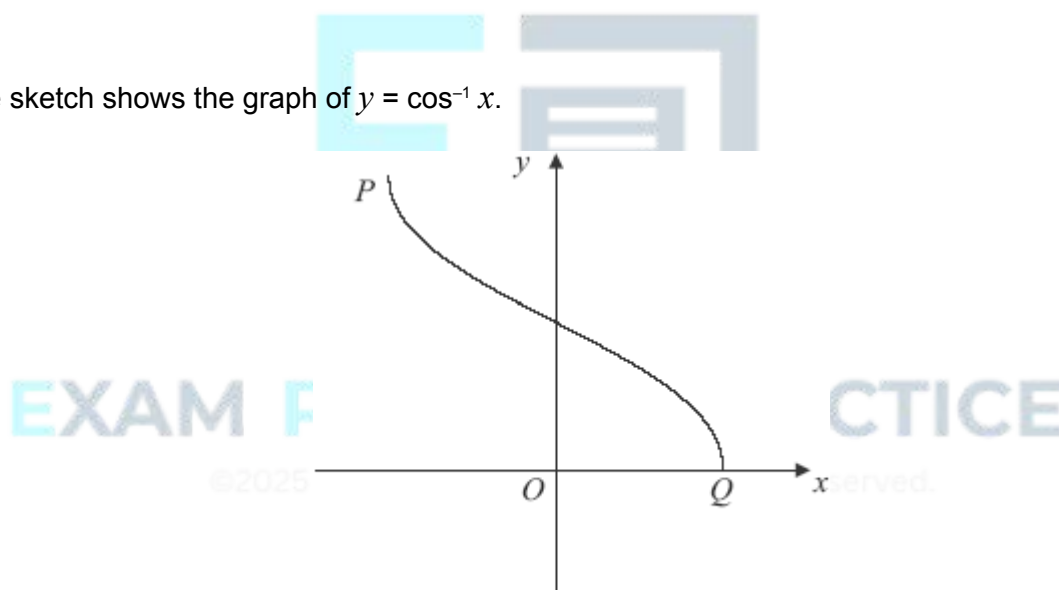
(c) Describe a single geometrical transformation that maps the graph of $y = 2x^{\frac{3}{2}}$ onto the graph of $y = 2(x + 3)^{\frac{3}{2}}$.

(2)

(Total 14 marks)

Q35.

The sketch shows the graph of $y = \cos^{-1} x$.



(a) Write down the coordinates of P and Q , the end points of the graph.

(2)

(b) Describe a sequence of two geometrical transformations that maps the graph of $y = \cos^{-1} x$ onto the graph of $y = 2 \cos^{-1}(x - 1)$.

(4)

(c) Sketch the graph of $y = 2 \cos^{-1}(x - 1)$.

(2)

(d) (i) Write the equation $y = 2 \cos^{-1}(x - 1)$ in the form $x = f(y)$.

(2)

- (ii) Hence find the value of $\frac{dx}{dy}$ when $y = 2$.

(3)

(Total 13 marks)



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